MATH 440/540 **HW 10** Schreyer

Due Thursday Oct. 24

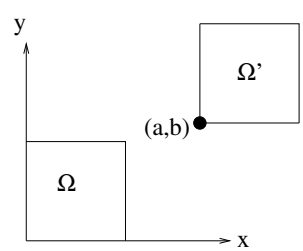
WITH SOLUTIONS

The following four problems provide more practice changing variables and looking at symmetries that harmonic functions have. Show that the coordinate transformation is well-defined (Jacobian). Then show (or not) that the Laplace equation does not change in the transformed coordinate system (change of variables, show that if $u_{xx} + u_{yy} = 0$ then $u_{x'x'} + u_{y'y'} = 0$ or equivalent in polar coordinates). Thus showing that if you have a harmonic function in Ω then doing a change of variables in certain cases gives you a harmonic function in Ω'.

1. (4 pts) Show that if $u(x, y)$ is a harmonic function in Ω, then $u(x+a,\ y+b)$ is harmonic in the domain

$$\Omega' = \{(x + a, y + b) \mid (x, y) \in \Omega\}.$$

A sample Ω and Ω' are given as:



This shows that if u is harmonic in a domain, and the domain is translated, then translating the coordinates gives a harmonic function in the new domain.

First let us check to see if the transformation is valid in the sense that we can go back and forth between the two coordinate systems without losing information:

$$J = \begin{vmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Since the Jacobian is not equal to 0, the transformation does not loose any information and we proceed!

We are given that $\triangle u(x, y) = 0$, and we want to show that $\triangle u(x', y') = 0$. Let us start

by calculating u_{xx} in terms of x' and y':

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial x} \\
&= u_{x'}(1) + u_{y'}(0) = u_{x'} \\
\frac{\partial^2 u}{\partial x^2} &= \frac{\partial u_x}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u_x}{\partial y'}\frac{\partial y'}{\partial x} \\
&= \frac{\partial u_x}{\partial x'}(1) + \frac{\partial u_x}{\partial y'}(0) \\
&= \frac{\partial}{\partial x'}(u_{x'}) = u_{x'x'}
\end{aligned}$$

Similarly, we get

$$\begin{aligned}
\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial y} \\
&= u_{x'}(0) + u_{y'}(1) = u_{y'} \\
\frac{\partial^2 u}{\partial y^2} &= \frac{\partial u_y}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u_y}{\partial y'}\frac{\partial y'}{\partial y} \\
&= \frac{\partial u_y}{\partial x'}(0) + \frac{\partial u_y}{\partial y'}(1) \\
&= \frac{\partial}{\partial y'}(u_{y'}) = u_{y'y'}
\end{aligned}$$

And so if we have $u_{xx} + u_{yy} = 0$ we also have

$$u_{x'x'} + u_{y'y'} = 0.$$

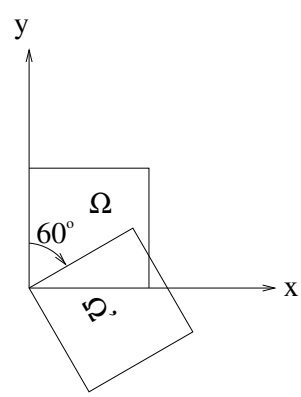
Thus if we have a harmonic function in a domain Ω, $u(x, y)$, we can get a harmonic function in the translated domain, Ω', by just doing a change of variables!

2. (4 pts) Show that if $u(x, y)$ is a harmonic function in Ω, then $u(x',\ y')$ is harmonic in the domain Ω', where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Omega' = \{(\cos(\alpha)x + y\sin(\alpha),\ -\sin(\alpha)x + \cos(\alpha)y) \mid (x, y) \in \Omega\}.$$

A sample Ω and Ω' for $\alpha = \pi/3$ are given as:



First let us check to see if the transformation is valid in the sense that we can go back and forth between the two coordinate systems without losing information:

$$J = \begin{vmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{vmatrix} = \begin{vmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{vmatrix} = 1$$

Since the Jacobian is not equal to 0, the transformation does not loose any information and we proceed.

We are given that $\triangle u(x,y) = 0$, and we want to show that $\triangle u(x',y') = 0$. Let us start by calculating u_{xx} in terms of x' and y':

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial x} \\
&= u_{x'}\cos(\alpha) + u_{y'}[-\sin(\alpha)] = \cos(\alpha)u_{x'} - \sin(\alpha)u_{y'} \\
\frac{\partial^2 u}{\partial x^2} &= \frac{\partial u_x}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u_x}{\partial y'}\frac{\partial y'}{\partial x} \\
&= \frac{\partial u_x}{\partial x'}\cos(\alpha) + \frac{\partial u_x}{\partial y'}[-\sin(\alpha)] \\
&= \cos(\alpha)\frac{\partial}{\partial x'}\left(\cos(\alpha)u_{x'} - \sin(\alpha)u_{y'}\right) - \sin(\alpha)\frac{\partial}{\partial y'}\left(\cos(\alpha)u_{x'} - \sin(\alpha)u_{y'}\right) \\
&= \cos^2(\alpha)u_{x'x'} - \cos(\alpha)\sin(\alpha)u_{x'y'} - \sin(\alpha)\cos(\alpha)u_{x'y'} + \sin^2(\alpha)u_{y'y'}
\end{aligned}$$

Similarly, we get

$$\frac{\partial^2 u}{\partial y^2} = \sin^2(\alpha)u_{x'x'} + \sin(\alpha)\cos(\alpha)u_{x'y'} + \cos(\alpha)\sin(\alpha)u_{x'y'} + \cos^2(\alpha)u_{y'y'}$$

And so if we have $u_{xx} + u_{yy} = 0$ we also have

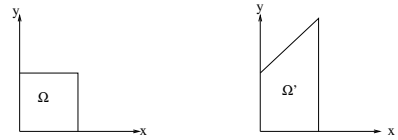
$$\begin{aligned}
0 &= u_{xx} + u_{yy} = \left[\cos^2(\alpha)u_{x'x'} - 2\cos(\alpha)\sin(\alpha)u_{x'y'} + \sin^2(\alpha)u_{y'y'}\right] \\
&\qquad + \left[\sin^2(\alpha)u_{x'x'} + 2\sin(\alpha)\cos(\alpha)u_{x'y'} + \cos^2(\alpha)u_{y'y'}\right] \\
&= u_{x'x'}\left[\cos^2(\alpha) + \sin^2(\alpha)\right] + u_{y'y'}\left[\sin^2(\alpha) + \cos^2(\alpha)\right] \\
&= u_{x'x'} + u_{y'y'}
\end{aligned}$$

Thus if we have a harmonic function in a domain Ω, $u(x,y)$, we can get a harmonic function in the rotated domain, Ω' by just doing a change of variables.

3. (4 pts) Show that if $u(x,y)$ is a harmonic function in Ω, then $u(x',\ y')$ is *not necessarily* harmonic in the domain Ω', where Ω and Ω' are in the left half plane of $I\!R^2$ $(x > 0)$:

$$x' = x^2 \qquad y' = x + y$$
$$\Omega' = \{(x^2,\ x+y) \mid (x,y) \in \Omega\}.$$

Hint: I got $\frac{\partial u}{\partial x} = u_{x'}(2x) + u_{y'}(1) = 2(x')^{1/2}u_{x'} + u_{y'}$. A sample Ω and Ω' are given as:



First let us check to see if the transformation is valid in the sense that we can go back and forth between the two coordinate systems without losing information:

$$J = \begin{vmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & 0 \\ 1 & 1 \end{vmatrix} = 2x$$

So the Jacobian is not zero in Ω as long as Ω is in the left or right half of the domain (i.e. Ω does not contain the y-axis).

We are given that $\triangle u(x,y) = 0$, and we want to show that $\triangle u(x',y') = 0$ but according to the problem statement this will not be the case. Let us see what happens. Start with calculating u_{xx} in terms of x' and y':

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial x} \\
&= u_{x'}(2x) + u_{y'}(1) = 2(x')^{1/2}u_{x'} + u_{y'} \\
\frac{\partial^2 u}{\partial x^2} &= \frac{\partial u_x}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u_x}{\partial y'}\frac{\partial y'}{\partial x} \\
&= \frac{\partial u_x}{\partial x'}(2x) + \frac{\partial u_x}{\partial y'}(1) \\
&= 2(x')^{1/2}\frac{\partial}{\partial x'}\left(2(x')^{1/2}u_{x'} + u_{y'}\right) + \frac{\partial}{\partial y'}\left(2(x')^{1/2}u_{x'} + u_{y'}\right) \\
&= 2(x')^{1/2}\left[(x')^{-1/2}u_{x'} + 2(x')^{1/2}u_{x'x'} + u_{x'y'}\right] + 2(x')^{1/2}u_{x'y'} + u_{y'y'} \\
&= 2u_{x'} + 4x'u_{x'x'} + 4(x')^{1/2}u_{x'y'} + u_{y'y'}
\end{aligned}$$

And (thankfully much easier), changing variables for u_{yy}:

$$\begin{aligned}
\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial y} \\
&= \frac{\partial u}{\partial x'}(0) + \frac{\partial u}{\partial y'}(1) = u_{y'} \\
\frac{\partial^2 u}{\partial y^2} &= \frac{\partial u_y}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u_y}{\partial y'}\frac{\partial y'}{\partial y} \\
&= u_{y'x'}(0) + u_{y'y'}(1) = u_{y'y'}
\end{aligned}$$

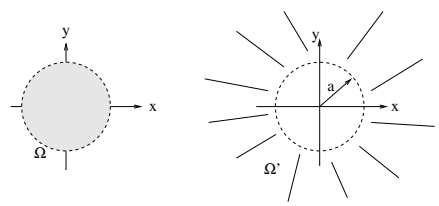
Thus

$$\triangle u = u_{xx} + u_{yy} = 4x'u_{x'x'} + 4(x')^{1/2}u_{x'y'} + 2u_{y'y'} + 2u_{x'} = 0$$

and unless u is a very special function, this does not imply that $u_{x'x'} + u_{y'y'} = 0$ (no symmetry with this coordinate transformation).

4. (4 pts) Show that if $u(r,\theta)$ is a harmonic function in a circle, $r^2 < a^2$, then $u(r',\ \theta')$ is harmonic in the exterior circle, $r^2 > a^2$, where

$$r'r = a^2 \qquad \theta' = \theta$$



$$J = \begin{vmatrix} \frac{\partial r'}{\partial r} & \frac{\partial r'}{\partial \theta} \\ \frac{\partial \theta'}{\partial r} & \frac{\partial \theta'}{\partial \theta} \end{vmatrix} = \begin{vmatrix} -\frac{a^2}{r^2} & 0 \\ 0 & 1 \end{vmatrix} = -\frac{a^2}{r^2}$$

Since the Jacobian is not equal to 0 in the domain, Ω, which is bounded, the transformation does not loose any information and we proceed! We start with $\triangle u = 0$ in the circle of radius a:

$$\triangle u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0.$$

Proceeding using the chain rule:

$$\begin{aligned}
\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial r'}\frac{\partial r'}{\partial r} + \frac{\partial u}{\partial \theta'}\frac{\partial \theta'}{\partial r} \\
&= \frac{\partial u}{\partial r'}\left(-\frac{a^2}{r^2}\right) + \frac{\partial u}{\partial \theta'}(0) \\
r\frac{\partial u}{\partial r} &= -\frac{a^2}{r}\frac{\partial u}{\partial r'} = -r'\frac{\partial u}{\partial r'} \\
\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) &= \frac{dr'}{dr}\frac{\partial}{\partial r'}\left(-r'\frac{\partial u}{\partial r'}\right) = \frac{a^2}{r^2}\frac{\partial}{\partial r'}\left(r'\frac{\partial u}{\partial r'}\right) \\
\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) &= \frac{a^2}{r^3}\frac{\partial}{\partial r'}\left(r'\frac{\partial u}{\partial r'}\right)
\end{aligned}$$

And knowing that since $\theta' = \theta$ we have $u_\theta = u_{\theta'}$ and $u_{\theta\theta} = u_{\theta'\theta'}$ we have

$$\begin{aligned}
\triangle u &= \frac{a^2}{r^3}\frac{\partial}{\partial r'}\left(r'\frac{\partial u}{\partial r'}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial (\theta')^2} \\
0 &= \frac{(r')^3}{a^4}\frac{\partial}{\partial r'}\left(r'\frac{\partial u}{\partial r'}\right) + \frac{(r')^2}{a^4}\frac{\partial^2 u}{\partial (\theta')^2}
\end{aligned}$$

Multiplying both sides of the last equation by $\dfrac{a^4}{(r')^4}$ (which we can do, because r' lives outside the circle in Ω' and so is never 0), we get:

$$0 = \frac{1}{r'}\frac{\partial}{\partial r'}\left(r'\frac{\partial u}{\partial r'}\right) + \frac{1}{(r')^2}\frac{\partial^2 u}{\partial (\theta')^2}$$

which shows that if we perform the change of variables on $u(r, \theta)$ (which is harmonic on the interior of a circle), we get a function, $u(r', \theta')$, that is harmonic on the exterior of the circle.

5. (16 pts) Z& T Chapter 7, 8.1. Prove that the set of functions $\cos(nx)$, $\sin(nx)$ (for $n = 1, 2, 3, ...$) and constants are orthogonal over the interval $[-\pi, \pi]$ by confirming the following 8 integral identities by hand:

   (a) $\int_{-\pi}^{\pi} \cos(nx)\, dx = 0$ (This shows that $\cos(nx)$ and constants are orthogonal).

   (b) $\int_{-\pi}^{\pi} \sin(nx)\, dx = 0$

   (c) $\int_{-\pi}^{\pi} \cos(nx)\sin(kx)\, dx = 0, \qquad k \neq n, \text{ and } k = n$

   (d) $\int_{-\pi}^{\pi} \cos(nx)\cos(kx)\, dx = 0, \qquad k \neq n$

   (e) $\int_{-\pi}^{\pi} \sin(nx)\sin(kx)\, dx = 0, \qquad k \neq n$

   (f) $\int_{-\pi}^{\pi} \cos^2(nx)\, dx = \pi$

   (g) $\int_{-\pi}^{\pi} \sin^2(nx)\, dx = \pi$

   (h) $\int_{-\pi}^{\pi} 1\, dx = 2\pi$

   Patience. Be careful not to divide by zero.

6. (6 pts) Z& T Chapter 7, 8.2. For the Fourier series over the interval $[-\pi, \pi]$:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

   show that

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx$$

   We want to derive the formula for the coefficients of $\sin(nx)$, b_n, in the Fourier series. Begin with the Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \big[a_n \cos(nx) + b_n \sin(nx)\big].$$

   We multiply the above equation by $\sin(kx)$ where k is a fixed positive integer, and integrate from $-\pi$ to π:

$$\int_{-\pi}^{\pi} f(x) \sin(kx)\, dx = \int_{-\pi}^{\pi} \frac{a_0}{2} \sin(kx)\, dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} \big(a_n \cos(nx)\sin(kx) + b_n \sin(nx)\sin(kx)\big) dx$$

The first integral on the right-hand side is zero. Let us assume that $f(x)$ is smooth enough so that the series converges uniformly and we can thus interchange the summation with the integral:

$$\begin{aligned}
\int_{-\pi}^{\pi} f(x)\sin(kx)\,dx &= \sum_{n=1}^{\infty}\left[a_n\int_{-\pi}^{\pi}\cos(nx)\sin(kx)\,dx + b_n\int_{-\pi}^{\pi}\sin(nx)\sin(kx)\,dx\right] \\
&= \sum_{n=1}^{\infty}[a_n(0)] + \sum_{n=1,n\neq k}^{\infty}[b_n(0)] + b_k\int_{-\pi}^{\pi}\sin(kx)\sin(kx)\,dx \\
&= b_k\pi
\end{aligned}$$

where we used the orthogonality conditions from the previous problem. Replacing k by n and solving for b_n we get the formula for calculating the coefficients for $\sin(nx)$:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

7. (4 pts) Z& T Chapter 7, 8.3 Show that the constant term in the FS of $f(x)$ is equal to the average of $f(x)$ over the interval $[-\pi,\pi]$, given by:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

Note: This is very useful for checking your calculation of a_0: If you can visualize $f(x)$ you can look at your value of $a_0/2$ and see if it makes sense. So for example, if your function is always positive over $[-\pi,\pi]$, a_0 should be positive as well!

Recall that the average of a function is the integral of the function over the interval, divided by the length of the interval. Thus

$$\text{average of } f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx\right] = \frac{1}{2}a_0.$$

8. (8 pts) Z& T Chapter 7, 8.4. Let $f(x)$ be sectionally continuous on $[-\pi,\pi]$.

   (a) If $f(x)$ is an even function show that its Fourier series is even and that its Fourier coefficients are given by

   $$a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos(nx)\,dx, \qquad n = 0,1,2,... \qquad b_n = 0, \qquad n = 1,2,3,...$$

   (b) If $f(x)$ is an odd function show that its Fourier series is odd and that its Fourier coefficients are given by

   $$a_n = 0, \qquad n = 0,1,2,... \qquad b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin(nx)\,dx, \qquad n = 1,2,3,...$$

Use the fact that if $f(x)$ is even, $f(-x) = f(x)$, and if $f(x)$ is odd, then $f(-x) = -f(x)$. If your work assumes, e.g. that an odd times and even function is odd, then this must be shown using the properties of odd and even functions.

We are given that f is sectionally continuous on $[-\pi, \pi]$. Let us suppose that f is an even function. What are it's Fourier coefficients? Consider them one at a time. We use the fact that $f(-x) = f(x)$ and that if we do a substitution, e.g. $u = -x$, then $du = -dx$:

$$\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{\pi}^{0} f(-x)\,(-1)dx + \int_{0}^{\pi} f(x)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(-x)\,dx + \int_{0}^{\pi} f(x)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right] \\
&= \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx
\end{aligned}$$

Using the fact that $\cos(nx)$ is an even function:

$$\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\cos(nx)\,dx + \int_{0}^{\pi} f(x)\cos(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{\pi}^{0} f(-x)\cos(-nx)\,(-1)dx + \int_{0}^{\pi} f(x)\cos(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(-x)\cos(-nx)\,dx + \int_{0}^{\pi} f(x)\cos(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(x)\cos(nx)\,dx + \int_{0}^{\pi} f(x)\cos(nx)\,dx\right] \\
&= \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos(nx)\,dx
\end{aligned}$$

Using the fact that $\sin(nx)$ is an odd function:

$$\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\sin(nx)\,dx + \int_{0}^{\pi} f(x)\sin(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{\pi}^{0} f(-x)\sin(-nx)\,(-1)dx + \int_{0}^{\pi} f(x)\sin(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(-x)\sin(-nx)\,dx + \int_{0}^{\pi} f(x)\sin(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\pi} f(x)(-1)\sin(nx)\,dx + \int_{0}^{\pi} f(x)\sin(nx)\,dx\right] \\
&= 0.
\end{aligned}$$

Finding the coefficients for an odd function is done similarly.

9. (22 pts) Z& T Chapter 7, 8.6. This is practice in writing a series so that there are only nonzero terms. Do NOT derive the Fourier series for (8.32) and (8.33). For all such problems, you can check by writing out the terms of your resulting series to see if you recover the nonzero terms of your original series.

(8.18) (warm up). Show that $\frac{\pi}{2} + \sum_{n=1}^{\infty} 2\frac{(-1)^n - 1}{\pi n^2}\cos(nx) = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos[(2k-1)x]}{(2k-1)^2}$.

(8.31) Now do similarly for this series - write it so that there are no zero terms. $\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n}\sin(nx)$

(8.32) (Same directions) $x(\pi - x) = \frac{\pi^2}{6} - 2\sum_{n=1}^{\infty}\frac{1+(-1)^n}{n^2}\cos(nx)$

(8.33) $x(\pi - x) = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^3}\sin(nx)$

(8.16) This one is a little tough. I recommend doing the other ones first.
$\frac{2+\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{-1+(-1)^n}{\pi n^2}\cos(nx) - \frac{1+(\pi-1)(-1)^n}{\pi n}\sin(nx)\right]$.

Compare (2 pts) Note that we have two different series for $x(\pi - x)$. How can this be? Maybe try graphing the first 3 terms of these two series (can use e.g. Desmos Graphing Calculator, or Geogebra, free online software that is easy to learn). No need to turn in graphs. Explain why we have two different series for the same function.

(8.18) We begin with the series:

$$\frac{\pi}{2} + \sum_{n=1}^{\infty} 2\frac{(-1)^n - 1}{\pi n^2}\cos(nx) = \frac{\pi}{2} + 2\frac{-2}{\pi}\cos(x) + 0 + 2\frac{-2}{9\pi}\cos(3x) + 0 + \ldots.$$

So the summation has nonzero values only when n is odd. Let $n = 2k-1$. Check: when $k = 1$, $n = 1$. When $k = 2$, $n = 3$. etc. We want to start the series when $n = 1$. This means we begin the series in terms of k at $k = 1$. We now have

$$\begin{aligned}
\frac{\pi}{2} + \sum_{k=1}^{\infty} 2\frac{(-1)^{2k-1} - 1}{\pi(2k-1)^2}\cos((2k-1)x) &= \frac{\pi}{2} + \sum_{k=1}^{\infty}\frac{-4}{\pi(2k-1)^2}\cos((2k-1)x) \\
&= \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos((2k-1)x)}{(2k-1)^2} \\
&= \frac{\pi}{2} - \frac{4}{\pi}\cos(x) - \frac{4}{9\pi}\cos(3x) + \ldots.
\end{aligned}$$

(8.31) We begin with the series:

$$\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n}\sin(nx) = \frac{2}{\pi}\left[2\sin(x) + 0\sin(2x) + \frac{2}{3}\sin(3x) + 0\sin(4x) + \frac{2}{5}\sin(5x) + ...\right]$$

So the summation has nonzero values only when n is odd. Let $n = 2k-1$. Check: when $k = 1$, $n = 1$. When $k = 2$, $n = 3$. etc. We want to start the series when $n = 1$. This means we begin the series in terms of k at $k = 1$. We now have

$$\begin{aligned}
\frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1-(-1)^{(2k-1)}}{2k-1}\sin((2k-1)x) &= \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{2}{2k-1}\sin((2k-1)x) \\
&= \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{1}{2k-1}\sin((2k-1)x) \\
&= \frac{4}{\pi}\left[\sin(x) + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + ...\right]
\end{aligned}$$

(8.32) Similarly, with the following series we use $n = 2k$:

$$\frac{\pi^2}{6} - 2\sum_{n=1}^{\infty}\frac{1+(-1)^n}{n^2}\cos(nx) = \frac{\pi^2}{6} - \sum_{k=1}^{\infty}\frac{1}{k^2}\cos(2kx)$$

(8.33) Similarly, with the following series we use $n = 2k-1$:

$$\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^3}\sin(nx) = \frac{8}{\pi}\sum_{k=1}^{\infty}\frac{1}{(2k-1)^3}\sin((2k-1)x)$$

(8.16) Grader: Ok to leave second term as it is (not necessary to break up into odd and even). If they don't break up odd and even, then both summations should really have the same index (k or n) but this is just for aesthetics, so don't take any points off.

This one is tough. I recommend doing the other ones first. We begin with the series:

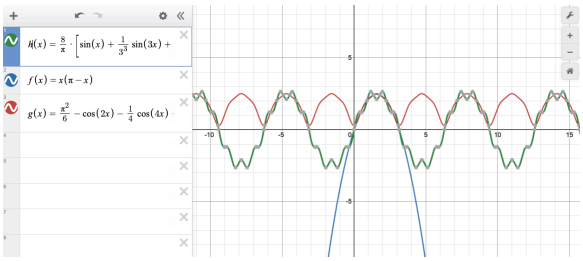
$$\begin{aligned}
\frac{2+\pi}{4} &+ \sum_{n=1}^{\infty}\left[\frac{-1+(-1)^n}{\pi n^2}\cos(nx) - \frac{1+(\pi-1)(-1)^n}{\pi n}\sin(nx)\right] \\
&= \frac{2+\pi}{4} + \frac{-2}{\pi}\cos(x) + \frac{-2}{9\pi}\cos(3x) - \frac{2-\pi}{\pi}\sin(x) - \frac{\pi}{2\pi}\sin(2x) - \frac{2-\pi}{3\pi}\sin(3x) + ...
\end{aligned}$$

The cos(nx) terms are zero for n *even* (note the mistake in the equation following (8.16) in the text book), and we have both odd and even nonzero terms for the $\sin(nx)$ terms. Breaking up the series into the nonzero odd terms, and then the nonzero even terms we get

$$\begin{aligned}
\frac{2+\pi}{4} &+ \sum_{k=1}^{\infty}\left[\frac{-2}{\pi(2k-1)^2}\cos[(2k-1)x] - \frac{2-\pi}{\pi(2k-1)}\sin[(2k-1)x]\right] \\
&- \sum_{k=1}^{\infty}\frac{1}{n}\sin(2kx)
\end{aligned}$$

Compare (2 pts) Note that we have two different series for $x(\pi - x)$. How can this be? Graphing the first 4 terms of these two series (see Figure), we see that (8.32) is an even extension, and (8.33) is an odd extension. They both do pretty well at capturing the function $x(\pi - x)$.

Blue curve is the original function $x(\pi - x)$.



Red curve is even extension (8.32). Green curve is odd extension (8.33).